DM – Recurrence relations

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Outline

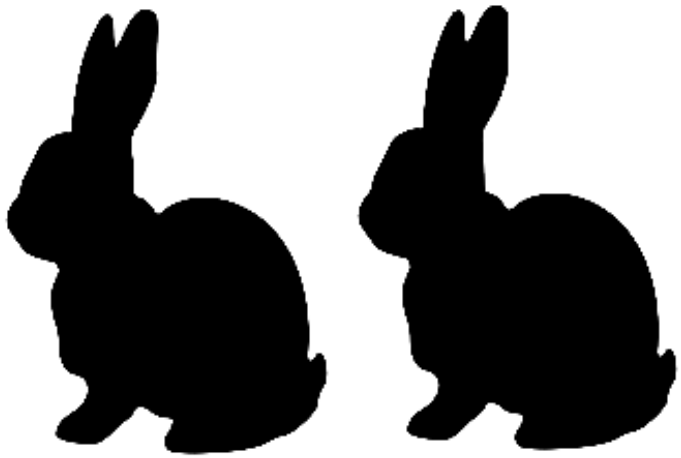
1. Examples of recurrence relations
2. Homogeneous linear recurrence relations
3. Non-homogeneous linear recurrence relations

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1. Examples of recurrence relations
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3. Non-homogeneous linear recurrence relations

Fibonacci sequence

A young pair of rabbits is placed on an island. A pair of rabbits does not breed until they are 2 months old. After they are 2 months old, each pair of rabbits produces another pair each month. Find a recurrence relation for the number of pairs of rabbits on the island after n months, assuming that no rabbits ever die.



Fibonacci sequence

New born **1-month old** **Adult**



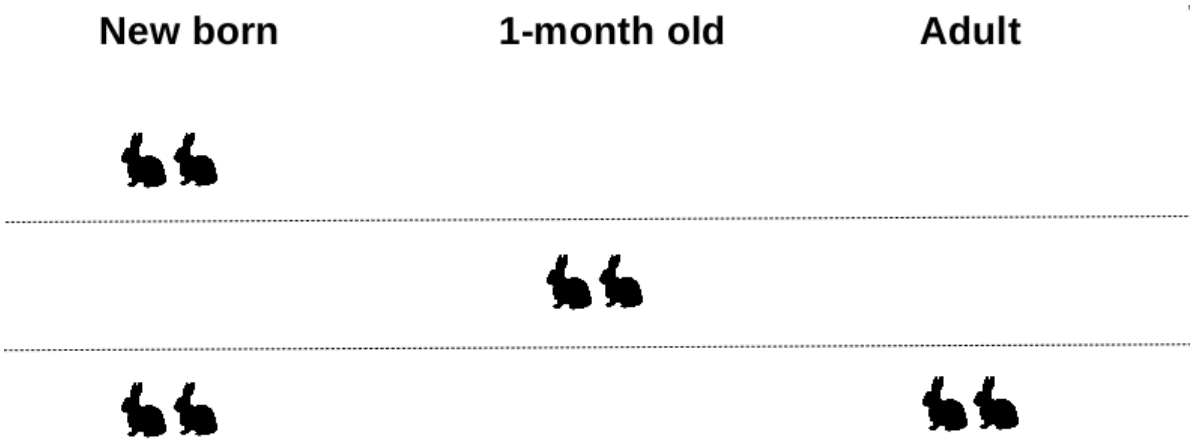
- $f_1 = 1$

Fibonacci sequence



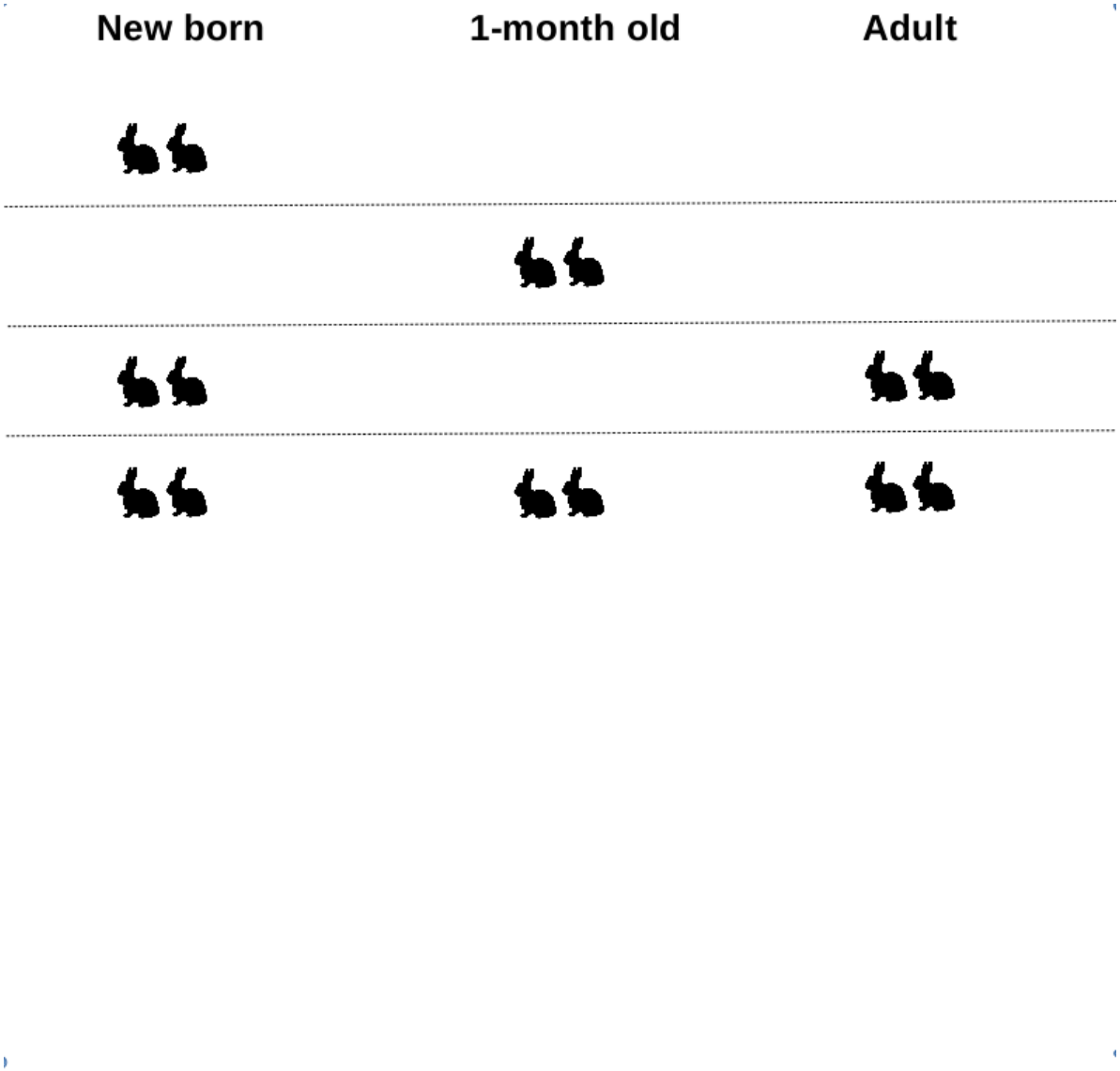
- $f_1 = 1$
- $f_2 = 1$

Fibonacci sequence



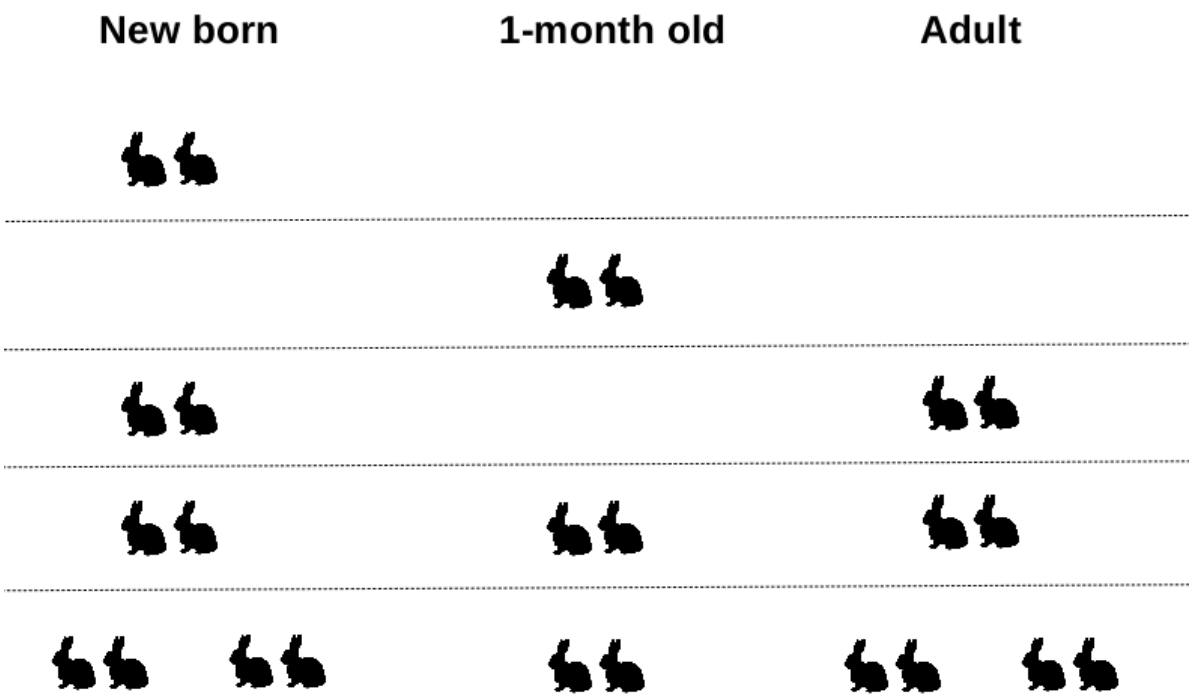
- $f_1 = 1$
- $f_2 = 1$
- $f_3 = 2$

Fibonacci sequence



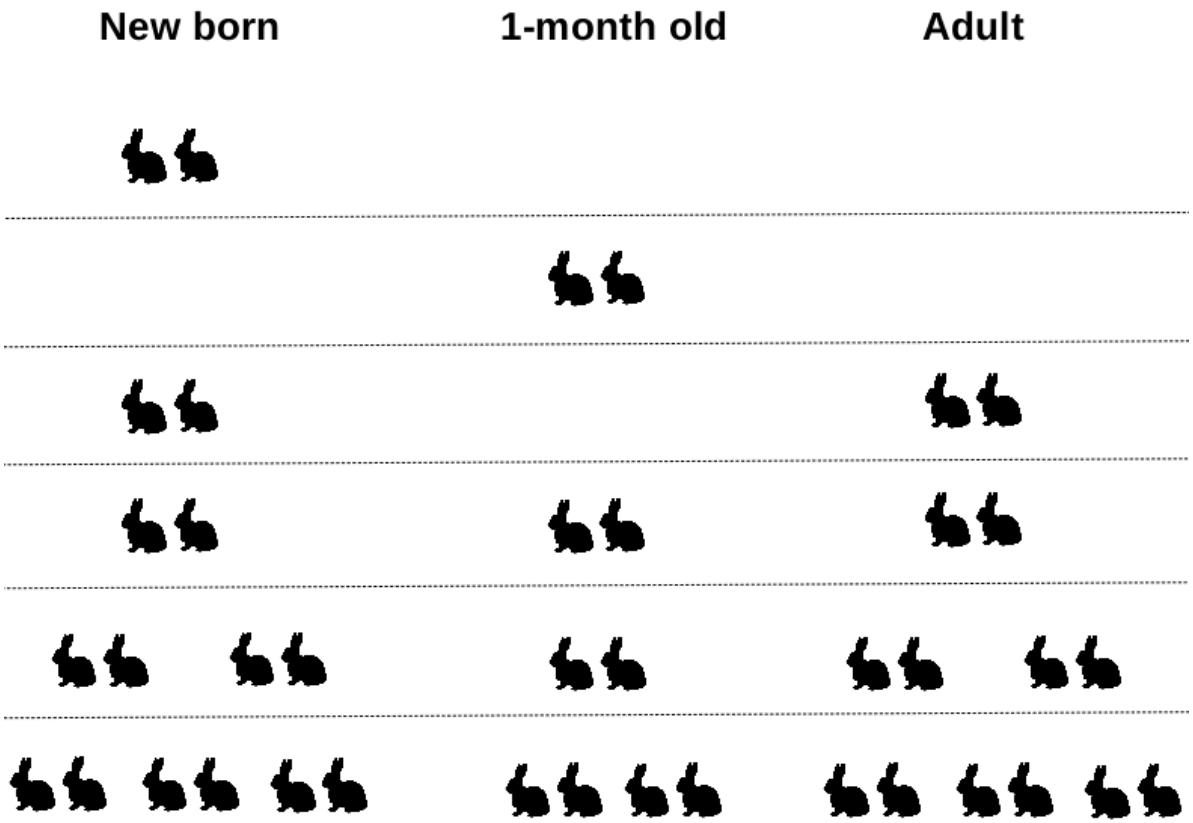
- $f_1 = 1$
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- $f_3 = 2$
- $f_4 = 3$

Fibonacci sequence



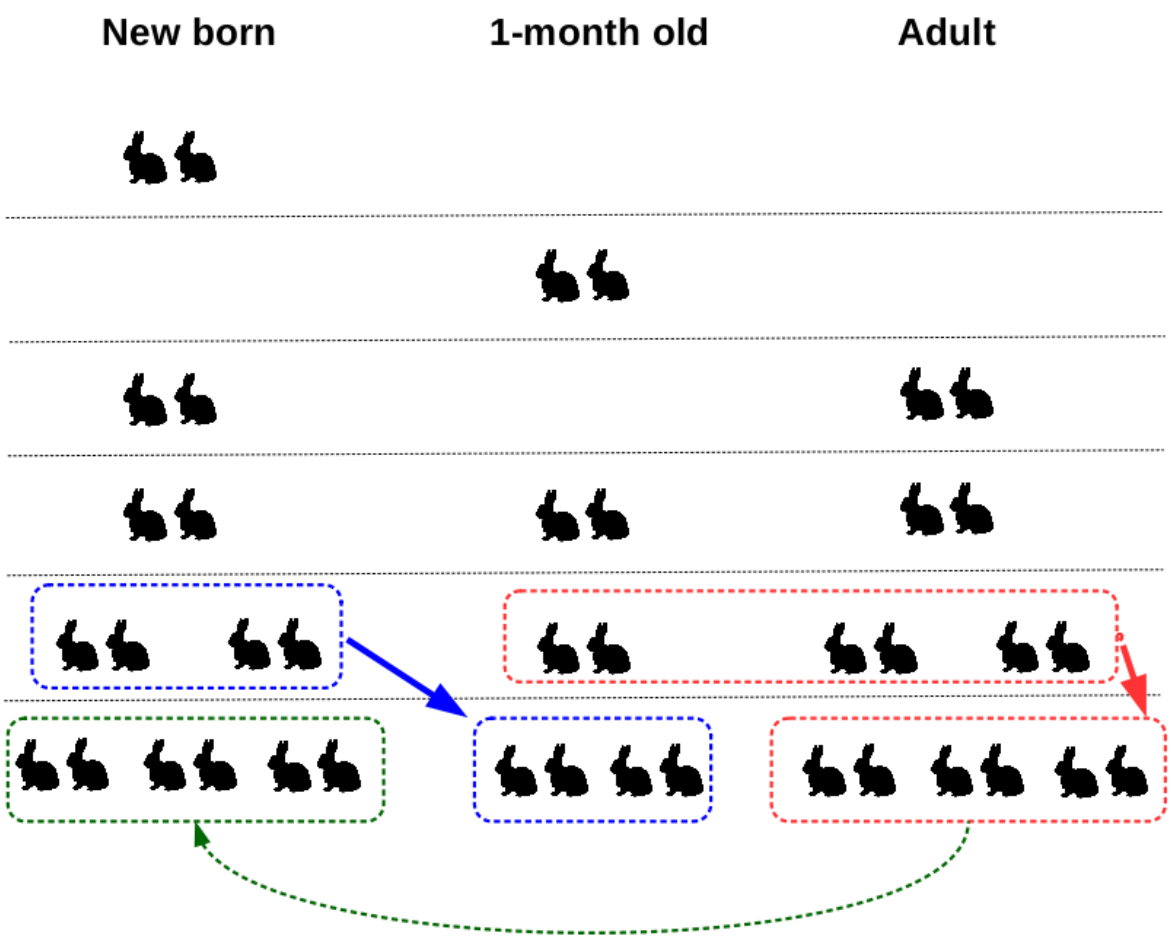
- $f_1 = 1$
- $f_2 = 1$
- $f_3 = 2$
- $f_4 = 3$
- $f_5 = 5$

Fibonacci sequence



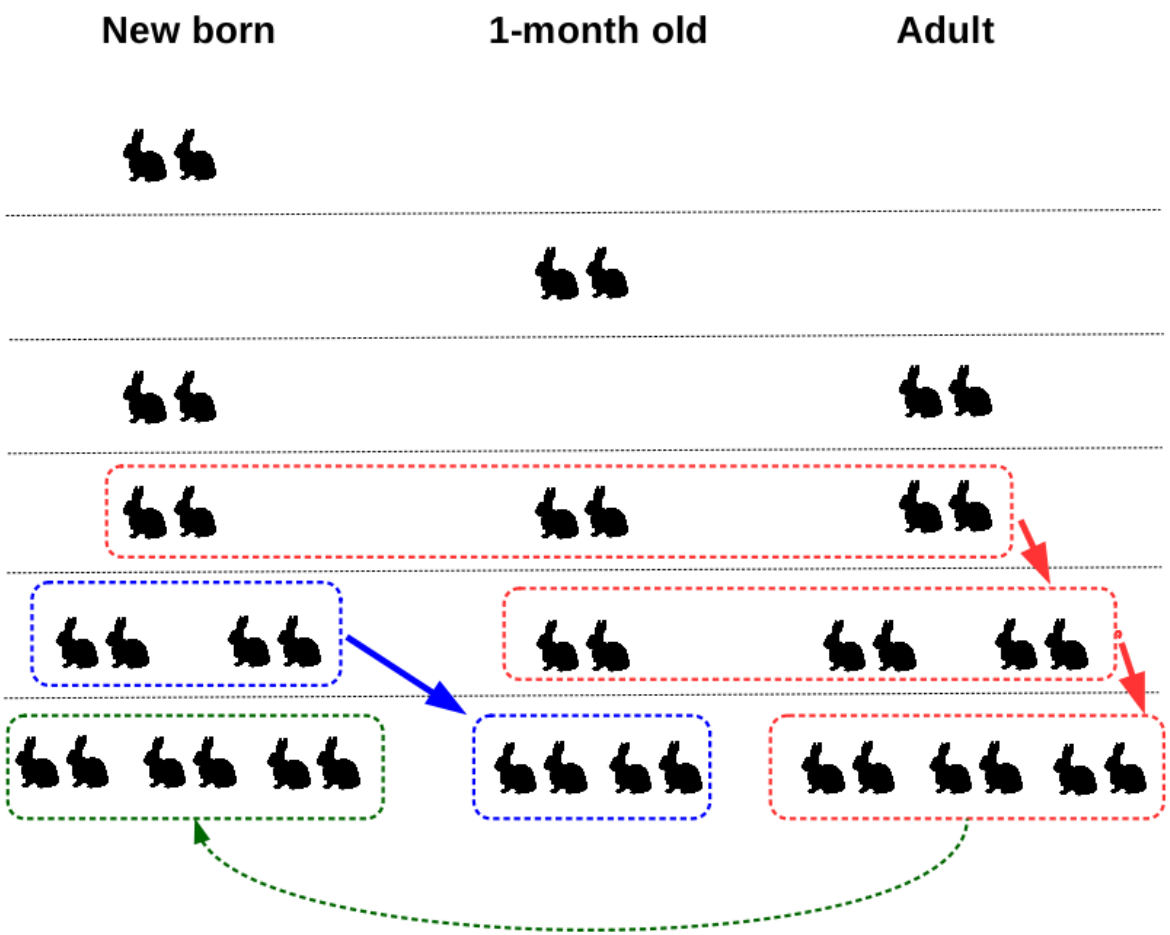
- $f_1 = 1$
- $f_2 = 1$
- $f_3 = 2$
- $f_4 = 3$
- $f_5 = 5$
- $f_6 = 8$

Fibonacci sequence



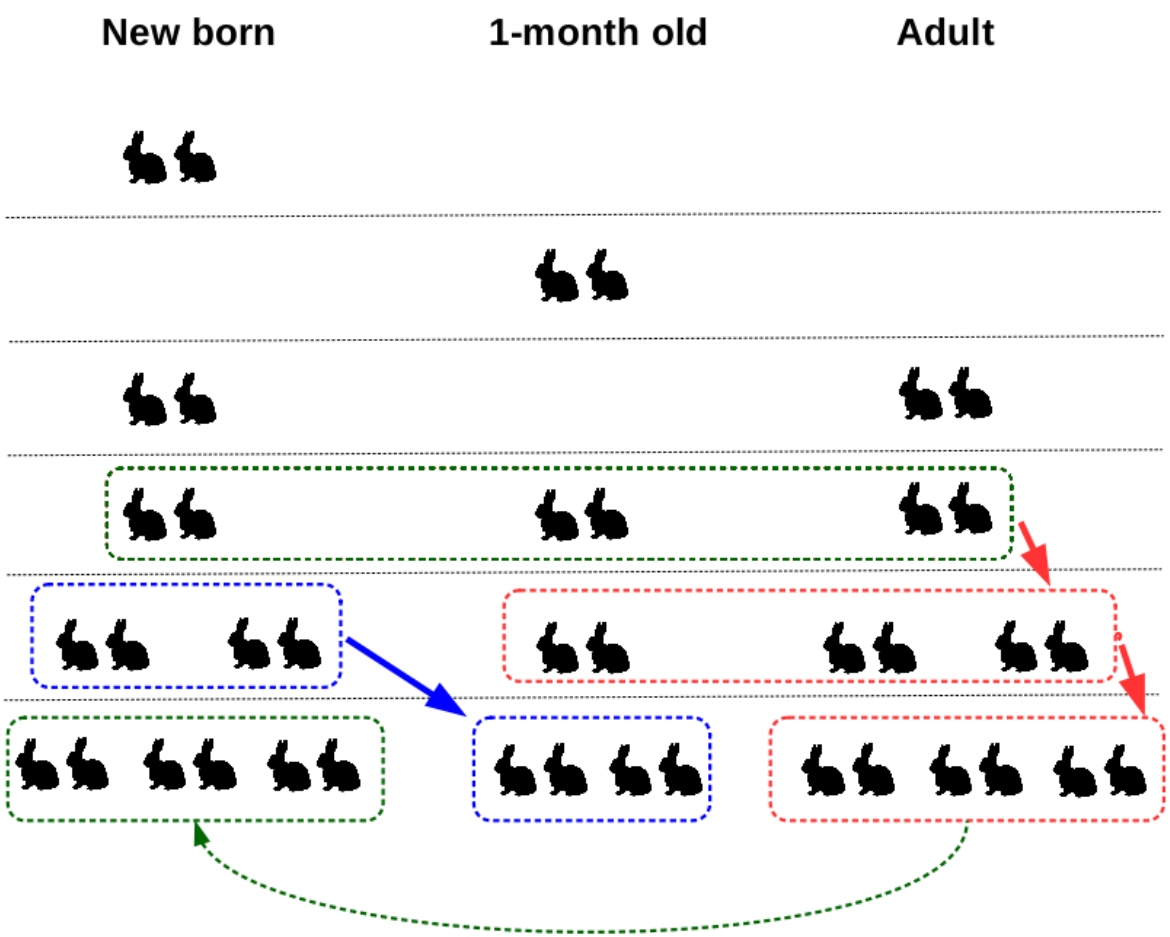
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Fibonacci sequence



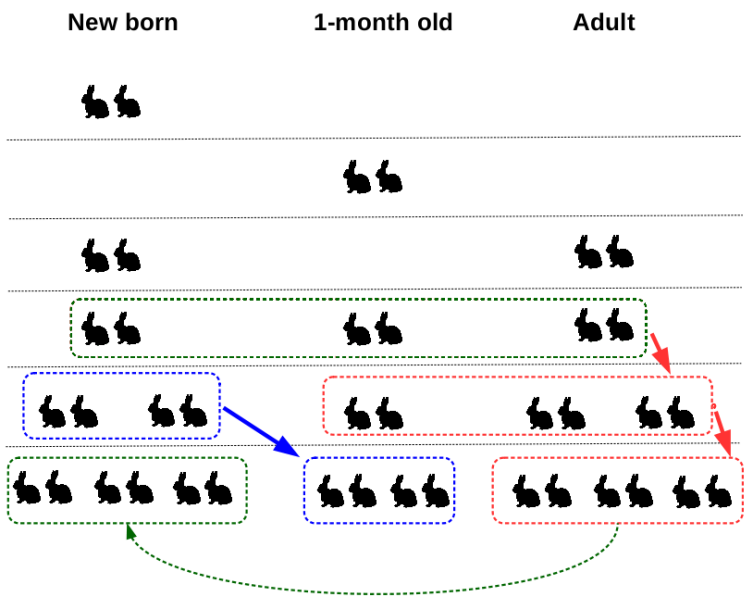
- $f_1 = 1$
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Fibonacci sequence



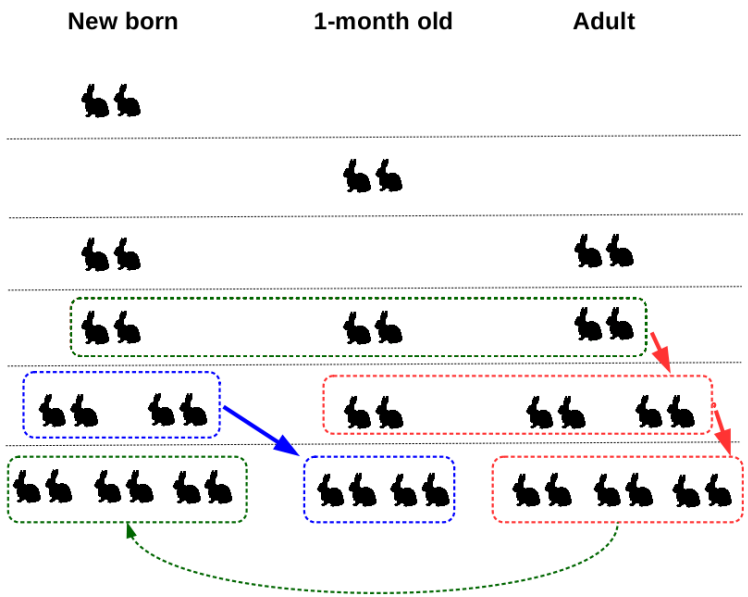
- $f_1 = 1$
- $f_2 = 1$
- $f_3 = 2$
- $f_4 = 3$
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- $f_6 = 8$

Fibonacci sequence



Key observation: The number of *newborn pairs* at month n is equal to the number of pairs at month $n-2$!

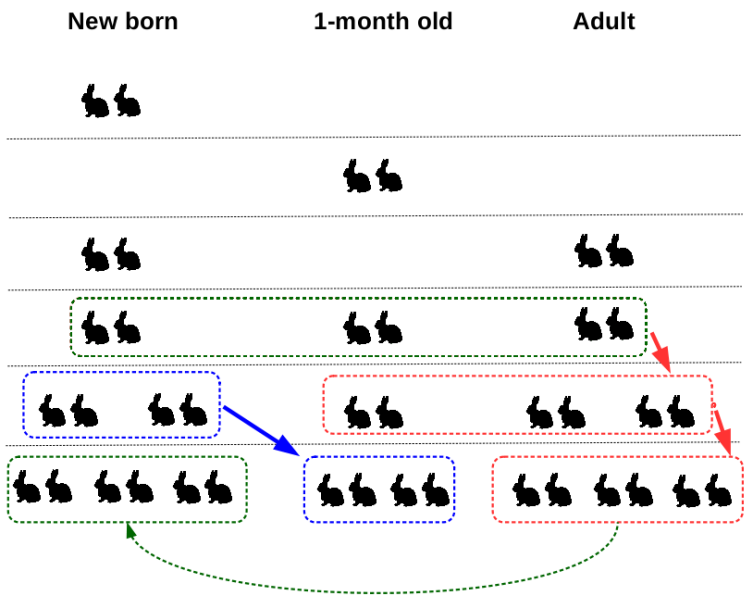
Fibonacci sequence



Key observation: The number of *newborn pairs* at month n is equal to the number of pairs at month $n-2$!

$$f_n - f_{n-1} = f_{n-2}$$

Fibonacci sequence



Key observation: The number of *newborn pairs* at month n is equal to the number of pairs at month $n-2$!

$$f_n = f_{n-1} + f_{n-2}\,, n \geq 3\,.$$

Fibonacci sequence

Definition

The sequence $(f_n)_{n\geq 1}$, *defined by the* recurrence relation

$$f_n = f_{n-1} + f_{n-2}\,, n \geq 3\,.$$

and the initial conditions $f_1 = f_2 = 1$ *is called the* Fibonacci sequence.

Remark: The initial conditions ensure that the sequence is *uniquely defined*.

Tower of Hanoi

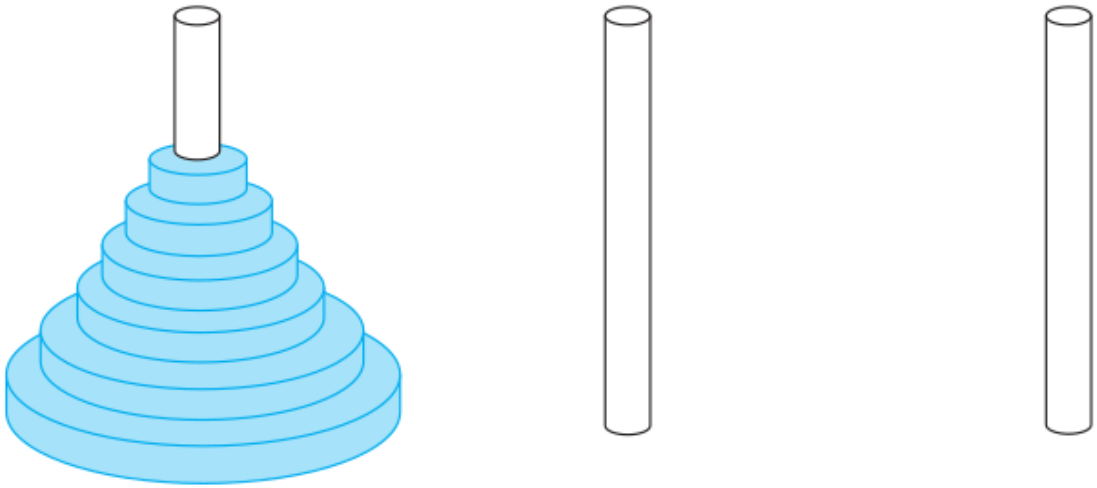


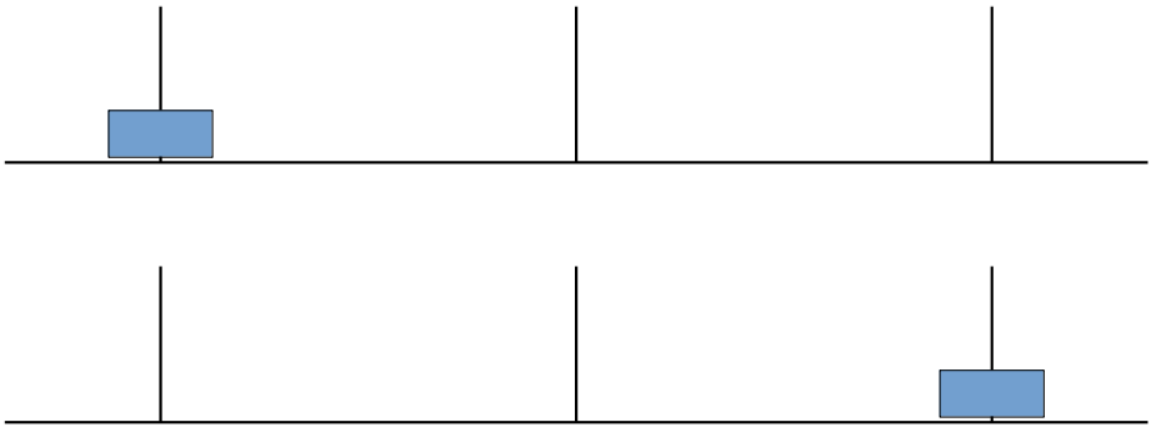
Figure: Rosen, p. 503

Rules:

- One disk moved at each step;
- A disk cannot be placed on top of a smaller one.

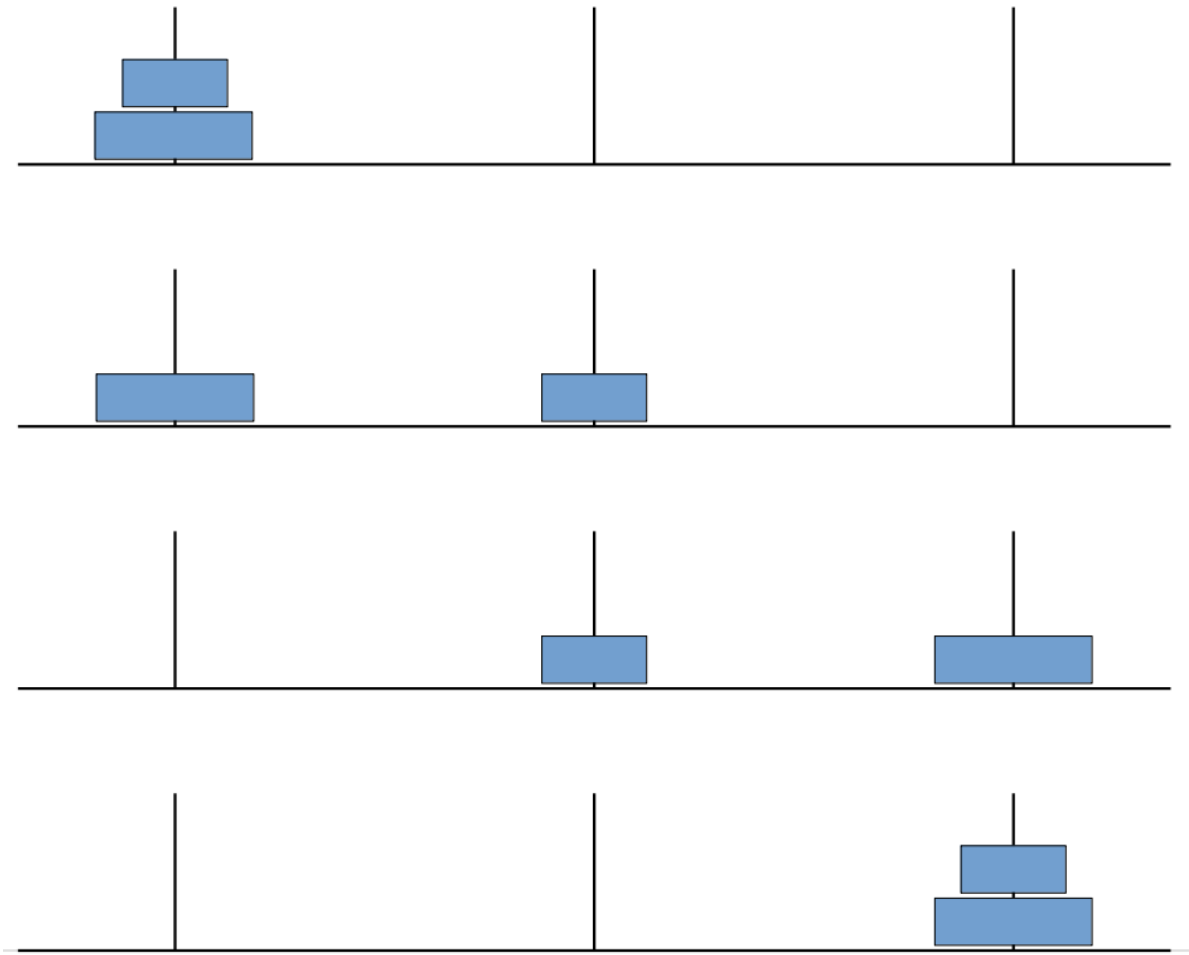
Goal: Move n disks from peg 1 to peg 3.

Tower of Hanoi



$$H_1 = 1$$

Tower of Hanoi

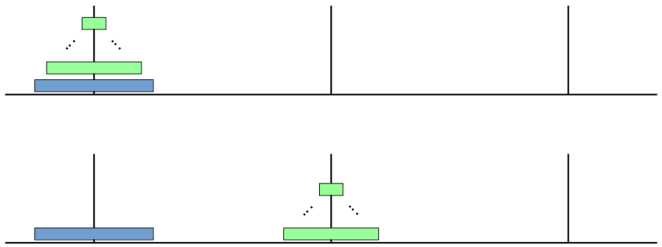


$H_2 = 3$

Tower of Hanoi

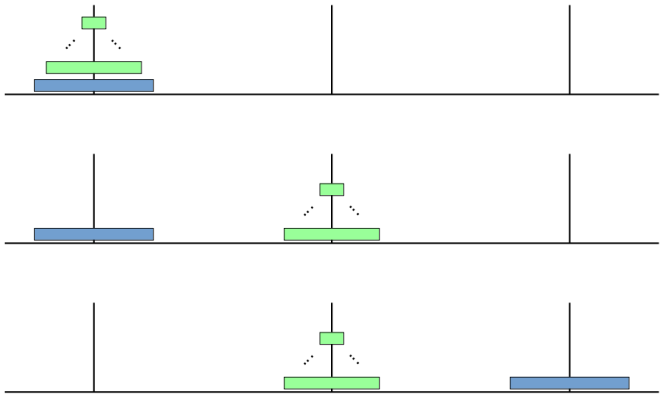


Tower of Hanoi



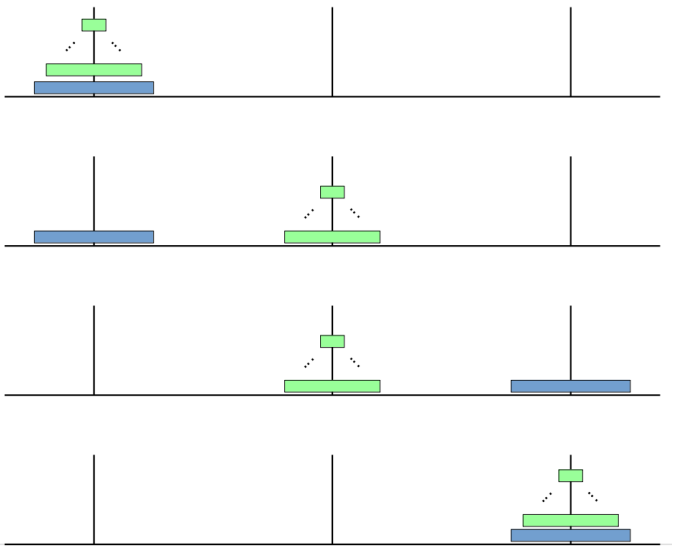
- Move $n-1$ top disks to peg 2: H_{n-1} steps;

Tower of Hanoi



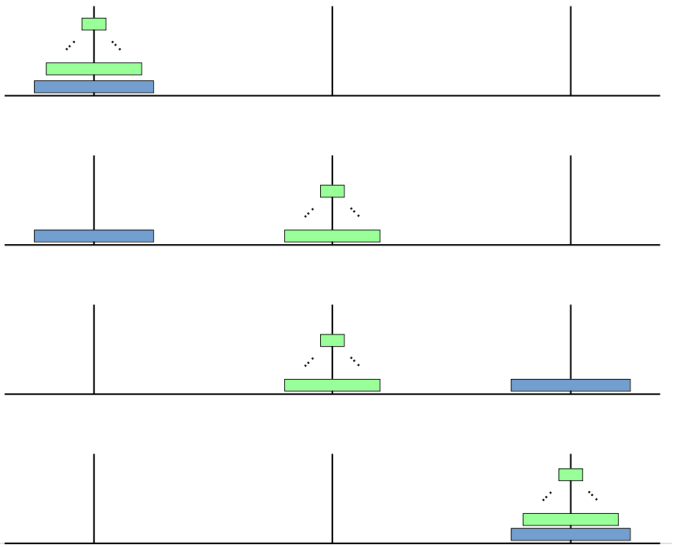
- Move $n-1$ top disks to peg 2: H_{n-1} steps;
- Move the largest disk to peg 3: 1 step;

Tower of Hanoi



- Move $n-1$ top disks to peg 2: H_{n-1} steps;
- Move the largest disk to peg 3: 1 step;
- Move $n-1$ disks from peg 2 to peg 3: H_{n-1} steps.

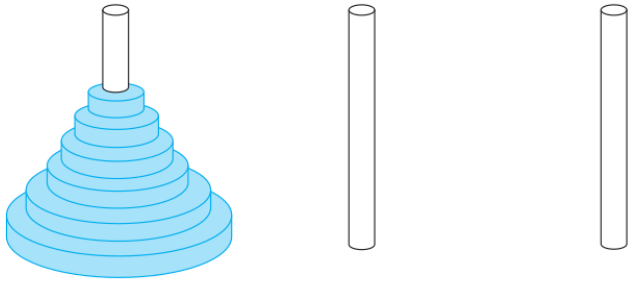
Tower of Hanoi



- Move $n-1$ top disks to peg 2: H_{n-1} steps;
- Move the largest disk to peg 3: 1 step;
- Move $n-1$ disks from peg 2 to peg 3: H_{n-1} steps.

$$H_n = H_{n-1} + 1 + H_{n-1} = 2H_{n-1} + 1\,.$$

Tower of Hanoi



$$\begin{cases} H_n = 2H_{n-1} + 1\,, n \geq 1\,, \\ H_1 = 1\,. \end{cases}$$

$H_1 = 1, H_2 = 3, H_3 = 7, H_4 = 15, \ldots$

General solution: $H_n = 2^n - 1$ for all $n \geq 1$ (induction on n).

For 64 disks, the number of moves is $2^{64} - 1 \approx 18 \times 10^{18}$.
At a rate of 1 move per second, without any break, it would take approximately 580 billion years to move 64 disks!

Valid codewords

A computer system considers a string of decimal digits a valid codeword if it contains an even number of 0 digits.

For instance, 1230407869 is valid, whereas 120987045608 is not valid.

Let a_n be the number of valid n-digit codewords. Find a recurrence relation for a_n.

The valid codewords of length 1 do not contain any zeroes:

$$1, 2, 3, 4, 5, 6, 7, 8, 9\,.$$

We have $a_1 = 9$.

The valid codewords of length 2 contain either none or two zeroes:

- Two zeroes: 00.
- No zeroes: 81 words (why?)

Hence $a_2 = 82$.

Valid codewords

In general, for a valid codeword w of length $n \geq 2$, either one of the following must hold:

Case 1: $w = cw'$ where

- c is a non-zero digit;
- w' contains an even number of zeroes.

There are 9 choices for c and a_{n-1} choices for w', hence $9a_{n-1}$ codewords of length n of this form.

Case 2: $w = 0w'$ where w' contains an odd number of zeroes.
There are $10^{n-1} - a_{n-1}$ such w' (why?), hence $10^{n-1} - a_{n-1}$ codewords of length n of this form.

Thus,

$$a_n = 9a_{n-1} + 10^{n-1} - a_{n-1} = 8a_{n-1} + 10^{n-1}\,, n \geq 2\,.$$

with initial condition $a_1 = 9$.

Outline

1. Examples of recurrence relations
2. Homogeneous linear recurrence relations
3. Non-homogeneous linear recurrence relations

Homogeneous linear recurrence relations

Definition

A homogeneous linear recurrence relation of degree k (with constant coefficients) is a recurrence linear relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} \tag{1}$$

where the c_i are real numbers and $c_k \neq 0$.

Remark: A sequence satisfying the relation (1) is uniquely determined by the *initial conditions*, ie. the first k terms $a_0, a_1, \ldots, a_{k-1}$.

Examples:

- The recurrence relation of the Fibonacci sequence that we have seen is linear and homogeneous.
- The recurrence relation of the Hanoi tower example is linear but not homogeneous.

Characteristic equation

Definition

The characteristic equation of the recurrence relation (1) is

$$x^k - c_1 x^{k-1} - \cdots - c_{k-1} x - c_k = 0 \tag{2}$$

where x is the unknown.

The roots of the equation (2) are called the characteristic roots of the relation (1).

Example: The characteristic equation of the Fibonacci sequence is

$$x^2 - x - 1 = 0\,.$$

This relation has 2 characteristic roots: $\frac{1\pm\sqrt{5}}{2}$.

Case $k = 2$

Theorem

Suppose that the equation (in x) $x^2 - c_1 x - c_2 = 0$ has two distinct *roots r_1 and r_2. Then the solutions to the relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ all have the form $a_n = \alpha r_1^n + \beta r_2^n$ where α, β are constants.*

Remark: The constants α, β are determined according to the initial conditions.

Example

Consider the Fibonacci sequence. The characteristic equation $x^2 - x - 1 = 0$ has two roots $r_{1,2} = \frac{1 \pm \sqrt{5}}{2}$.

The general term of the sequence will have the form

$$f_n = \alpha \left(\frac{1+\sqrt{5}}{2} \right)^n + \beta \left(\frac{1-\sqrt{5}}{2} \right)^n$$

In order to obtain α, β, let $n = 0, 1$ and solve the following system:

$$\begin{cases} \alpha + \beta = f_0 = 0 \\ \left(\frac{1+\sqrt{5}}{2} \right) \alpha + \left(\frac{1-\sqrt{5}}{2} \right) \beta = f_1 = 1 \end{cases}$$

We get $\alpha = \frac{1}{\sqrt{5}}, \beta = -\frac{1}{\sqrt{5}}$, and

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n .$$

Case $k = 2$

Theorem

Suppose that the characteristic equation has only one root r_0. Then the solutions to the relation have the form $a_n = \alpha r_0^n + \beta n r_0^n$.

Example: Solve $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1, a_1 = 6$.

The characteristic equation $x^2 - 6x + 9 = 0$ has only one root $r = 3$, hence

$$a_n = \alpha 3^n + \beta n 3^n .$$

Using the initial conditions, we get

$$\begin{cases} \alpha = a_0 = 1 \\ 3\alpha + 3\beta = a_1 = 6 \end{cases}$$

which yield $\alpha = \beta = 1$.

Hence $a_n = 3^n + n3^n$.

General case

Theorem

Suppose that the equation (2) has k distinct roots $r_1, r_2, \ldots, r_k$. Then the solution to the relation (1) has the form

$$a_n = \alpha_1 r_1^n + \cdots + \alpha_k r_k^n ,$$

where the constants α_i are determined by the initial conditions.

General case

Theorem

Suppose that the equation (2) has t roots $r_1, r_2, \ldots, r_t$ with multiplicities $m_1, \ldots, m_t$ ($m_i \geq 1$ and $m_1 + \cdots + m_t = k$) Then the solution to the relation (1) has the form

$$a_n = p_1(n)r_1^n + \cdots + p_t(n)r_t^n$$

where each p_i is a polynomial of degree not exceeding $m_i - 1$ and whose coefficients are determined by the initial conditions.

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Non-homogeneous linear recurrence relations

Definition

A non-homogeneous linear recurrence relation of degree k (with constant coefficients) is a recurrence linear relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n) \tag{3}$$

where the c_i are real numbers, $c_k \neq 0$ and $F(n)$ is a non-zero function in n.

The recurrence relation (1) is calleds the associated homogeneous recurrence relation of (3).

Theorem

If $(a_n^{(p)})$ is a particular solution to the non-homogeneous relation (3), then every solution of (3) has the form $(a_n^{(p)} + a_n^{(h)})$, where $(a_n^{(h)})$ is a solution to the associated homogeneous relation.

Example

Consider the relation $a_n = 3a_{n-1} + 2n$ with $a_1 = 3$.

The associated homogeneous relation $a_n = 3a_{n-1}$ has solutions $a_n^{(h)} = \alpha 3^n$ where α is a constant.

One can verify that $a_n^{(p)} = -n - \frac{3}{2}$ is a particular solution.

Hence the general solutions have the form

$$a_n = -n - \frac{3}{2} + \alpha 3^n \,.$$

Let $n = 1$ and use the initial condition $a_1 = 3$, we get $-1 - \frac{3}{2} + 3\alpha = 1$, whence $\alpha = \frac{11}{6}$.

The solution is $a_n = -n - \frac{3}{2} + \frac{11}{6} 3^n$.

Particular solution

Theorem

Suppose that $F(n)$ has the form $F(n) = Q(n)s^n$, where Q is a polynomial of degree t and s is a real number.

1. *If s is* not *a root of the characteristic equation of the associated homogeneous relation, then there is a particular solution of the form $p(n)s^n$, where p is a polynomial of degree not exceeding t.*
2. *If s is a root of multiplicity m of the characteristic equation of the associated omogeneous relation, then there is a particular solution of the form $n^m p(n)s^n$, where p is a polynomial of degree not exceeding t.*

Example

Consider $a_n = 1 + 2 + \cdots + n$. This sequence satisfies the relation

$$a_n = a_{n-1} + n\,,\, a_1 = 1\,.$$

The associated homogeneous relation is $a_n = a_{n-1}$, whose solution is $a_n = c$ (constant).
We have $F(n) = n\ 1^n$, and $s = 1$ is a single root of the characteristic equation of the associated relation, so we look for a particular solution of the form $a_n^{(p)} = n^1(p_0 + p_1 n)\ 1^n = p_0 n + p_1 n^2$.
Plugging this into the relation:

$$np_0 + n^2 p_1 = (n-1)p_0 + (n-1)^2 p_1 + n\,,$$

hence $(2p_1 - 1)n + (p_0 - p_1) = 0$, hence $2p_1 - 1 = p_0 - p_1 = 0$, and $p_0 = p_1 = \frac{1}{2}$. We have found a particular solution $a_n^{(p)} = \frac{n(n+1)}{2}$.
Thus, $a_n = \frac{n(n+1)}{2} + c$. Using $a_1 = 1$, we get $c = 0$.
The final solution is $a_n = \frac{n(n+1)}{2}$.